PUNJAB ENGINEERING COLLEGE (DEEMED TO BE UNIVERSITY)
Sector 12, Chandigarh - 160012 | Office of Dean Academic Affairs (DAA)
Capstone Coursework Deliverable • Academic Clearance Automation Engine

Data Flow Diagrams (DFD) & Data Dictionary

Academic Grievance, No-Dues & Clearance Automation Engine

Institution: Punjab Engineering College (Deemed to be University), Sector 12, Chandigarh - 160012
Document Identifier: PEC-SE-DFD-2026-FROZEN
Standard Compliance: Structured Systems Analysis & Design (Yourdon & DeMarco / Gane & Sarson Standards)
Baseline Version: 3.0.0 (FROZEN ARCHITECTURAL BASELINE)
Administrative Authority: Office of the Dean Academic Affairs (DAA), PEC Chandigarh
Status: FROZEN & RATIFIED (Implementation-Agnostic Baseline)

Baseline Freeze Declaration

> **NOTICE OF BASELINE FREEZE:**
> This document specifies the complete, immutable data transformations, data store structures, and information flow boundaries for the PEC Clearance & Grievance Automation Engine.
>
> The process hierarchies, data store schemas, input/output data flows, and data element dictionaries defined herein are **FROZEN**. Implementation changes (e.g., frontend state management, database engine replacement, or network transport layers) preserve these exact inputs, transformations, and outputs, ensuring zero modifications to this document.

1. Notational Conventions & Symbol Guide

Symbol Element	Gane & Sarson / Yourdon Notation	Mermaid Representation	Semantic Meaning
External Entity	Double Square / Rectangle	`Entity["External Entity"]`	Source or destination of data outside system boundaries (e.g., Student, Staff).
Process	Rounded Rectangle / Circle	`Process(("ID: Process Name"))`	Transforms incoming data flows into outgoing data flows.
Data Store	Open Rectangle / Parallel Lines	`Store[("DS-ID: Data Store")]`	Repository of persistent data at rest (e.g., relational tables).
Data Flow	Directed Vector Arrow	`A -- "Data Flow Name" --> B`	Pipeline carrying packetized data elements between entities, processes, and stores.

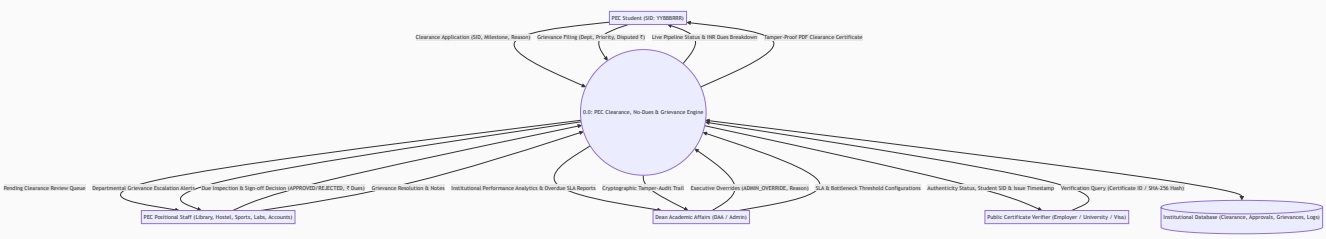
1.1 Formal DFD Construction Rules & Quality Compliance

This specification adheres strictly to the classic **Yourdon & DeMarco** and **Gane & Sarson** structured systems analysis rules:

Structured Analysis Rule	Rule Description & Formal Constraint	Compliance Verification in this DFD
Rule 1: Conservation of Data	A process cannot create data out of nothing ("Miracle") nor consume data without producing output ("Black Hole").	Every process (`1.0` through `6.0` and `2.1` .. `5.4`) transforms identifiable inputs into outputs.
Rule 2: Entity-to-Entity Forbidden	External entities cannot communicate directly within the DFD boundary without passing through a process.	Verified: All student, staff, and admin interactions pass through numbered processes.
Rule 3: Store-to-Store Forbidden	Data cannot move directly from one data store to another without process manipulation.	Verified: Table updates (e.g. `DS-4` to `DS-3`) are mediated by processes `3.0` , `5.1` , and `5.2` .
Rule 4: Entity-to-Store Forbidden	External entities cannot directly read or write to data stores without an intervening process.	Verified: All persistence calls are strictly routed through processes `1.0` , `2.0` , `3.0` , `4.0` .
Rule 5: Balancing Rule (Horizontal & Vertical)	Inputs and outputs of a parent process at Level N must exactly match the aggregate boundary flows at Level $N+1$.	Verified: Level 0 external flows balance with Level 1, and Level 1 processes balance with Level 2.
Rule 6: Unique Hierarchical Numbering	Every process is uniquely identified using hierarchical decimal notation (`0.0` $\rightarrow$ `1.0` .. `6.0` $\rightarrow$ `2.1` .. `2.4`).	Verified: Clear hierarchical numbering applied across all levels.

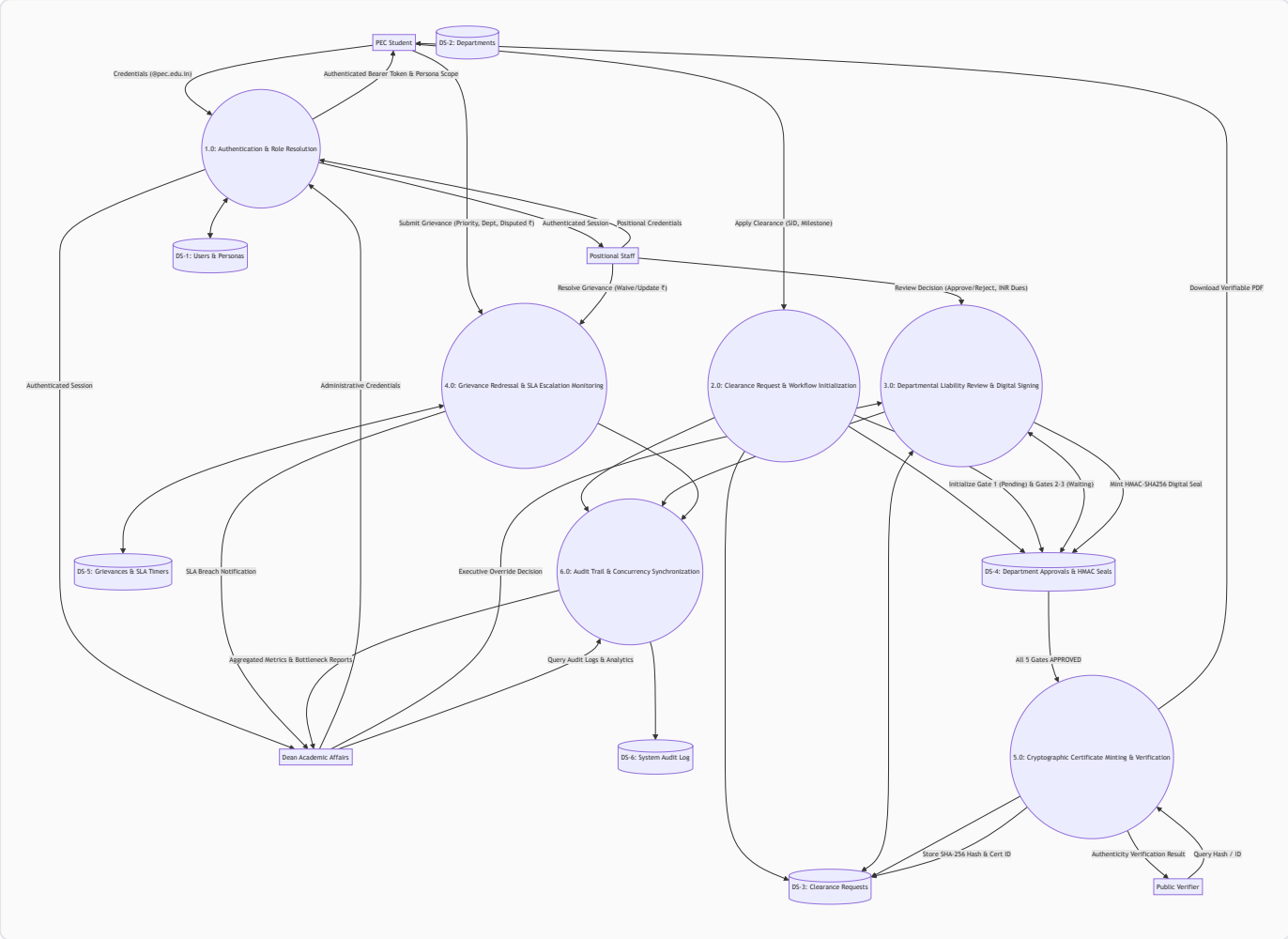
2. DFD Level 0: Context Analysis Diagram

The Level 0 Context Diagram establishes the highest-level operational boundary of the Clearance Engine within the Punjab Engineering College ecosystem. It identifies the four primary external entities interacting with the central clearance processing environment:



3. DFD Level 1: System Decomposition Diagram

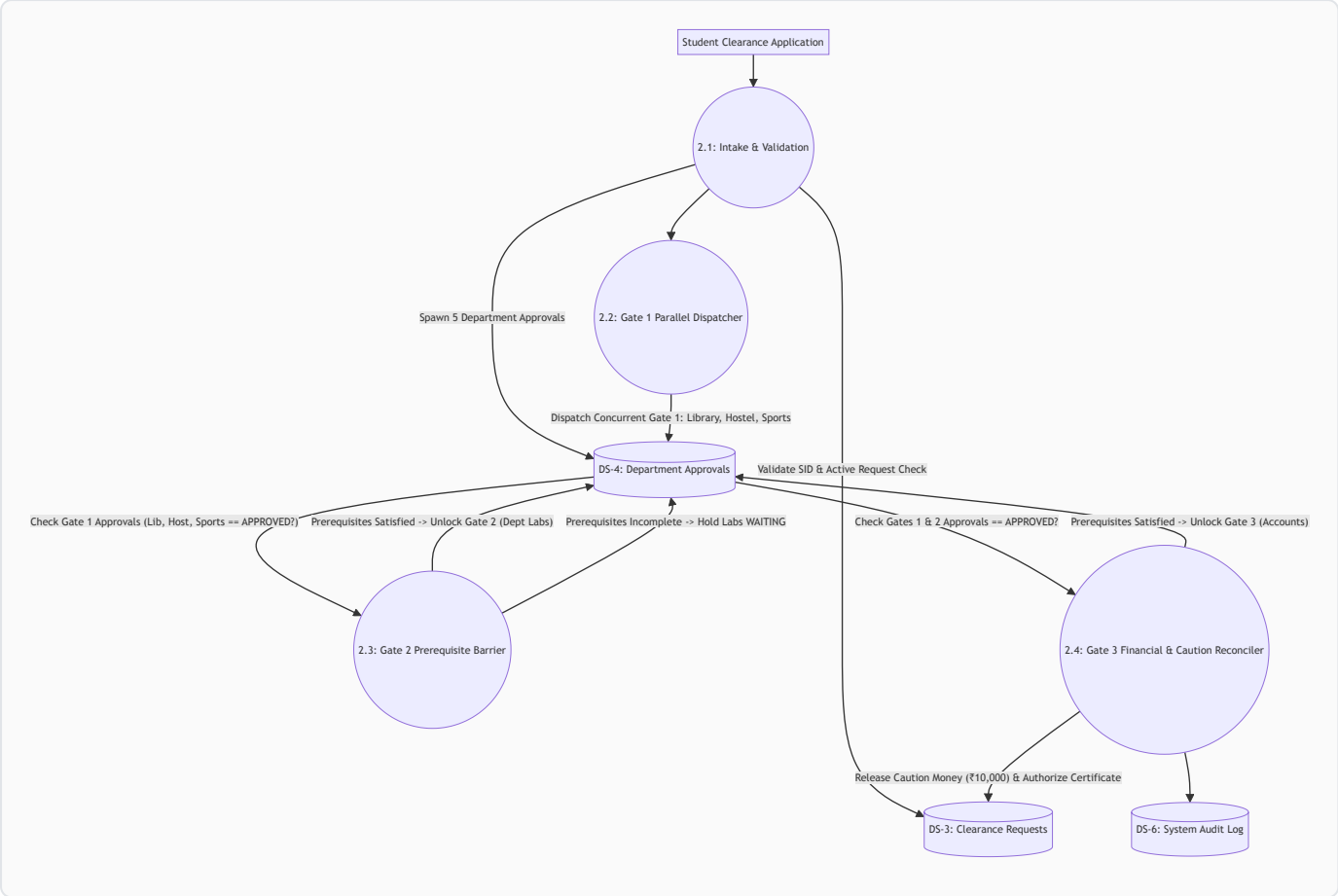
The Level 1 DFD decomposes the top-level process `0.0` into six modular, cohesive functional processes governing authentication, clearance workflow, liability assessment, grievance arbitration, cryptographic certification, and audit synchronization:



4. DFD Level 2: Sub-Process Decompositions

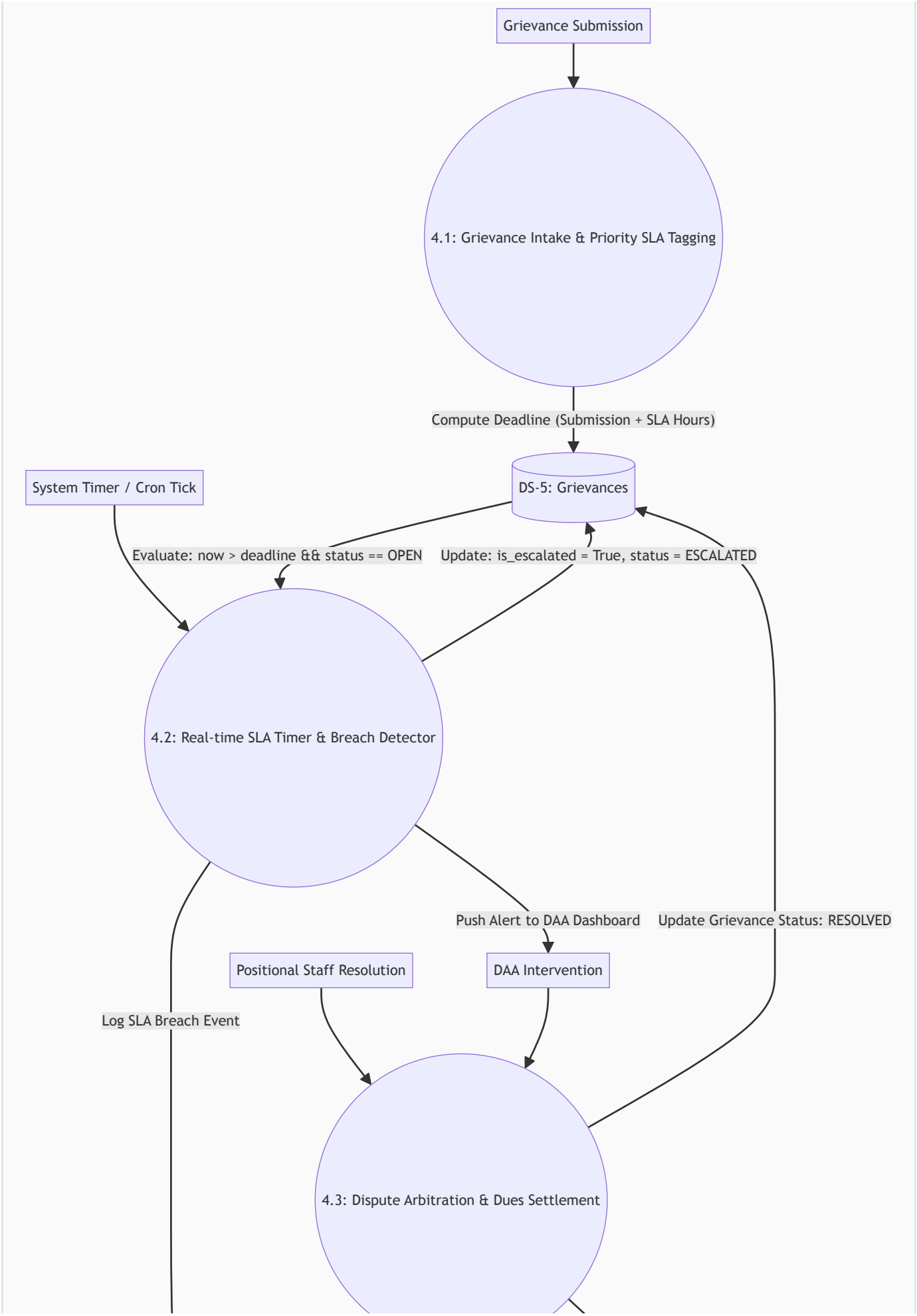
4.1 Process 2.0 Decomposition: Clearance Gating & Chain of Responsibility

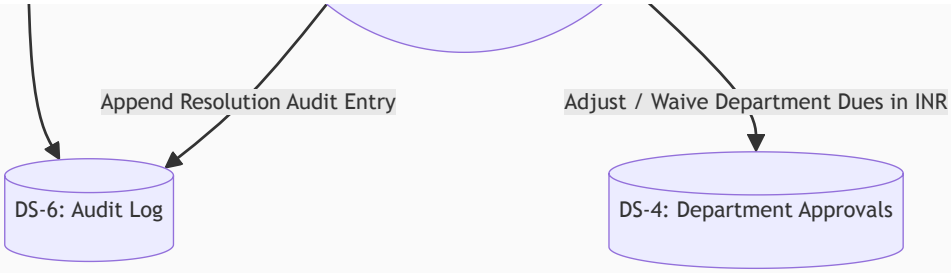
This diagram isolates how departmental clearance requests flow across the three sequential and parallel validation gates:



4.2 Process 4.0 Decomposition: Grievance Redressal & SLA Escalation Engine

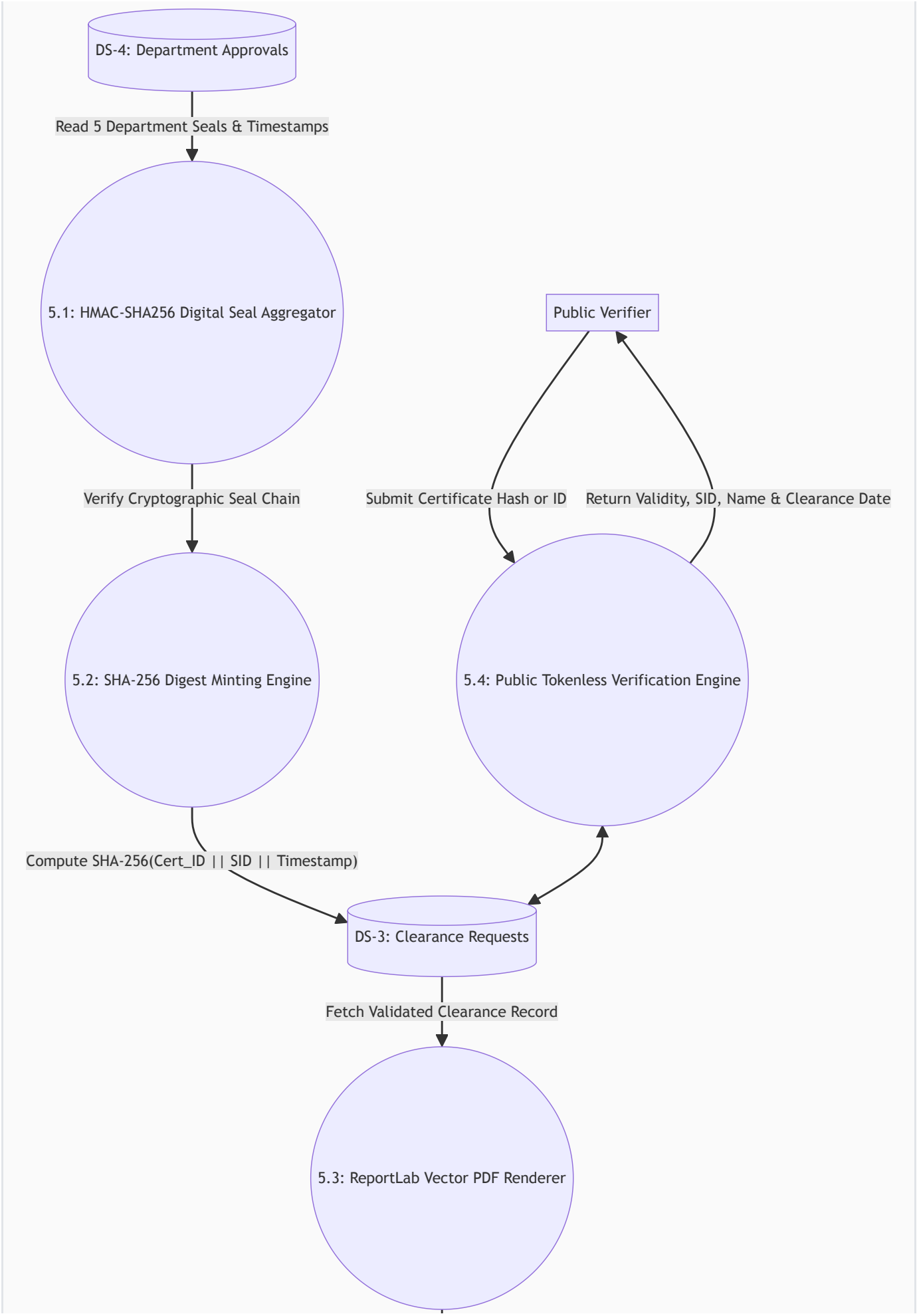
This diagram details the priority-based SLA calculation, automatic timer evaluation, and DAA escalation workflow:

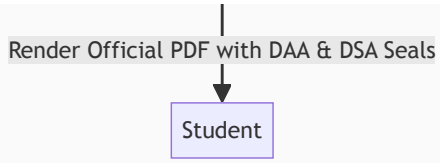

Grievance Submission
4.1: Grievance Intake & Priority SLA Tagging
Compute Deadline (Submission + SLA Hours)
DS-5: Grievances
System Timer / Cron Tick
Evaluate: now > deadline && status == OPEN
Update: is_escalated = True, status = ESCALATED
4.2: Real-time SLA Timer & Breach Detector
Push Alert to DAA Dashboard
Update Grievance Status: RESOLVED
Positional Staff Resolution
DAA Intervention
Log SLA Breach Event
4.3: Dispute Arbitration & Dues Settlement



4.3 Process 5.0 Decomposition: Cryptographic Certificate Minting & Public Verification

This diagram details the generation of HMAC digital signatures, SHA-256 certificate hashing, and unauthenticated public verification:


DS-4: Department Approvals
Read 5 Department Seals & Timestamps
5.1: HMAC-SHA256 Digital Seal Aggregator
Verify Cryptographic Seal Chain
5.2: SHA-256 Digest Minting Engine
Compute SHA-256(Cert_ID || SID || Timestamp)
Public Verifier
Submit Certificate Hash or ID
Return Validity, SID, Name & Clearance Date
5.4: Public Tokenless Verification Engine
DS-3: Clearance Requests
Fetch Validated Clearance Record
5.3: ReportLab Vector PDF Renderer



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5. Comprehensive Data Dictionary

5.1 Data Store Schemas (Data at Rest)

`DS-1` : Users & Personas (`users` table)

Field Name	Type	Key	Description	Example
`id`	Integer	PK	Unique user record surrogate key	`1`
`email`	String(120)	Unique	Institutional PEC email address	`alice@pec.edu.in`
`full_name`	String(120)	-	Student or Positional Staff Name	`Alice Sharma` / `Chief Warden (Hostels)`
`role`	String(20)	-	Role scope (`STUDENT` , `STAFF` , `ADMIN`)	`STUDENT`
`sid`	String(20)	Nullable	8-digit PEC Student Identification Number	`22103042`
`department_code`	String(20)	Nullable	Assigned department for staff	`HOSTEL`
`is_active`	Boolean	-	Account active status flag	`True`

`DS-2` : Departments (`departments` table)

Field Name	Type	Key	Description	Example
`id`	Integer	PK	Department record identifier	`2`
`code`	String(20)	Unique	Institutional department code	`HOSTEL`
`name`	String(120)	-	Official facility designation	`PEC Hostels (Shivalik/Kurukshetra/etc.)`
`head_title`	String(120)	-	Permanent administrative title	`Chief Warden (Hostels)`
`gate_number`	Integer	-	Pipeline gate assignment (`1` , `2` , or `3`)	`1`

`DS-3` : Clearance Requests (`clearance_requests` table)

Field Name	Type	Key	Description	Example
`id`	Integer	PK	Clearance request identifier	`1`
`student_id`	Integer	FK	References `users.id`	`1`
`academic_year`	String(20)	-	Academic session	`2025-2026`
`semester`	Integer	-	Current enrolled semester	`8`
`reason`	String(120)	-	Clearance justification	`B.Tech Final Degree Graduation`
`status`	String(30)	-	Overall status (`PENDING` , `APPROVED` , `REJECTED`)	`APPROVED`
`certificate_id`	String(50)	Unique	Minted institutional certificate number	`PEC-CLR-2026-00042`
`certificate_hash`	String(64)	Unique	Cryptographic SHA-256 certificate digest	`e3b0c44298fc1c149afbf4c8996fb92427ae41e4649b`

`DS-4` : Department Approvals (`department_approvals` table)

Field Name	Type	Key	Description	Example
`id`	Integer	PK	Approval gate record key	`1`
`request_id`	Integer	FK	References `clearance_requests.id`	`1`
`department_id`	Integer	FK	References `departments.id`	`2`
`status`	String(30)	-	`WAITING_PREREQUISITE`, `PENDING`, `APPROVED`, `REJECTED`	`APPROVED`
`due_amount`	Float	-	Outstanding department liability in INR (`₹`)	`3500.0`
`remarks`	Text	-	Positional reviewer notes	`Shivalik Mess Dues Reconciled`
`digital_signature`	String(128)	-	Cryptographic HMAC-SHA256 signature seal	`a8f9c1d2e3b4a5...`
`signed_at`	DateTime	-	Timestamp of cryptographic signing	`2026-10-15 14:32:00`

`DS-5`: Grievances (`grievances` table)

Field Name	Type	Key	Description	Example
`id`	Integer	PK	Grievance ticket identifier	`1`
`student_id`	Integer	FK	References `users.id`	`1`
`department_id`	Integer	FK	References `departments.id`	`2`
`subject`	String(200)	-	Concise grievance description	`Incorrect Mess Dues for Internship Term`
`priority`	String(20)	-	SLA Priority (`CRITICAL`, `HIGH`, `MEDIUM`, `LOW`)	`HIGH`
`disputed_amount`	Float	-	Contested liability in INR (`₹`)	`3500.0`
`status`	String(20)	-	`OPEN`, `RESOLVED`, `REJECTED`, `ESCALATED`	`ESCALATED`
`sla_deadline`	DateTime	-	Dynamic computed resolution deadline	`2026-10-16 02:30:00`
`is_escalated`	Boolean	-	Flag indicating DAA SLA breach escalation	`True`

`DS-6`: Audit Log (`audit_logs` table)

Field Name	Type	Key	Description	Example
`id`	Integer	PK	Sequential audit event identifier	`101`
`user_id`	Integer	FK	Actor responsible for the state mutation	`4`
`action`	String(50)	-	Standard action verb	`APPROVAL_SIGNED`, `ADMIN_OVERRIDE`
`entity_type`	String(50)	-	Target domain entity	`ClearanceRequest`, `Grievance`
`details`	Text	-	JSON-encoded mutation metadata	`{"dept": "HOSTEL", "status": "APPROVED"}`
`timestamp`	DateTime	-	UTC/IST creation timestamp	`2026-10-15 14:32:01`

5.2 Core Data Element Dictionary (Data in Motion)

1. **`SID` (Student Identification Number):**
   - **Type:** String (8 characters alphanumeric).
   - **Format:** `YYBBBRRR` where `YY` is enrollment year, `BBB` is academic branch code (`103` = CSE), and `RRR` is roll index.
   - **Scope:** Immutable institutional identifier.
2. **`due_amount_inr` (Financial Liability):**
   - **Type:** IEEE 754 Floating Point (formatted as `en-IN` decimal string in client).
   - **Currency:** Indian Rupee (`₹`).
   - **Allowable Range:** $[0.00, ; 500000.00]$.
3. **`digital_signature` (HMAC Seal):**
   - **Type:** Hexadecimal string (64 characters).
   - **Algorithm:** $\text{HMAC-SHA256}(K_{\text{institutional}}, \text{staff_id} \parallel \text{dept_code} \parallel \text{request_id} \parallel \text{timestamp})$.
4. **`certificate_hash` (Verification Digest):**
   - **Type:** Hexadecimal string (64 characters).
   - **Algorithm:** $\text{SHA-256}(\text{cert_id} \parallel \text{student_sid} \parallel \text{academic_year} \parallel \text{completion_time})$.
5. **`sla_priority`:**
   - **Type:** Enumeration (`CRITICAL`, `HIGH`, `MEDIUM`, `LOW`).
   - **Mapping:** `CRITICAL` $\rightarrow 4\text{h}$, `HIGH` $\rightarrow 12\text{h}$, `MEDIUM` $\rightarrow 24\text{h}$, `LOW` $\rightarrow 48\text{h}$.